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Treating differently or equally: A study exploring attitudes towards AI moral advisors

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ABSTRACT

Artificial intelligence (AI) technology has evolved from serving primarily as a decision-maker in the past to increasingly taking on the role of an advisor. However, contemporary attitudes toward AI applications in moral decision-making remain unclear. In Study 1, we explored whether there is a difference in attitudes towards human and AI moral advisors when both are defined in the context of a one-time decision-making scenario. Studies 2a and 2b, with the goal of achieving higher ecological validity, optimized decision-making methods and scenarios, respectively. We obtained consistent results, indicating that people equally trust the advice of AI moral advisors and human moral advisors. When it comes to assigning responsibility after a decision, individuals assign responsibility equally to both AI and human advisors.

1. Introduction

With the rapid advancement of Artificial Intelligence (AI) technology, its capabilities and the range of applications continue to expand. It is increasingly employed to assist or even directly take over human tasks. AI systems now collaborate with humans in various domains, from artistic creation (Zhou & Lee, 2024) to decision-making in human resources and clinical settings (Lee et al., 2021; Park et al., 2021). Given its superior accuracy and efficiency, specific tasks are delegated to specialized AI systems, such as analyzing vast amounts of data in large datasets to derive systematic answers from the data. Although AI often demonstrates advantages in many scenarios compared to human workers, a critical factor in determining whether it can fulfill its role and realize its value may lie in users' attitudes towards AI.

Across different domains, users show inconsistent attitudes towards AI. Dietvorst and his collaborators (2015) first introduced the concept of algorithm aversion, which means that, despite evidence showing that algorithmic predictions consistently outperform human predictors, people tend to avoid using algorithms after discovering their flaws, and the loss of trust in them is faster than in human experts. Some studies have indicated people's aversion to algorithms. For instance, individuals perceive algorithmic decisions as less fair and less trustworthy than human decisions. When tasks require unique human skills rather than mechanical computational abilities, negative emotions are more likely to be triggered (Lee, 2018). Moreover, even if the medical services provided by AI are as effective as those provided by humans, consumers are unwilling to use AI for healthcare decision-making in real or hypothetical scenarios, whether evaluated individually or jointly. This is because people believe AI will overlook the uniqueness of the consumer. However, when AI is used for personalized medical diagnoses or merely to assist doctors in making diagnoses, rather than acting as an independent decision-maker, this effect is diminished (Longoni et al., 2019). Algorithm aversion is more likely to occur when users have incorrect expectations or when AI recommendations contradict their intuition (Burton et al., 2020). The factors that may trigger algorithm aversion depend on the specific decision task and domain in which AI is involved (Castelo et al., 2019).

However, some experimental results indicate an opposite attitude: algorithm appreciation. People perceive AI performance to be at least as good as human experts, and, in certain cases, they may trust AI more than human experts, such as predicting song popularity or romantic attractiveness. In a series of experiments, it was found that participants, across various domains, placed greater importance on algorithmic estimates compared to human estimates, whether in individual or joint evaluations (Logg et al., 2019).

The above studies suggest that people's attitudes toward AI decision-

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0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

making are not stable, possibly due to different contexts leading to varied outcomes. The current research focuses on the context of moral decision-making, with AI playing the role of an advisor providing recommendations. Through this, we aim to explore how individuals approach artificial intelligence in this specific context.

2. Attitudes toward AI ethical decision-making

People may differ on what constitutes "correct" behavior and whether there is an objective truth in ethics (Graham, 2010; Harman, 1975; Mackie, 1990). For instance, Kant (1785/1997) emphasizes the intentions of the agent, while Bentham (1781) and Mill (1998) lean toward focusing more on outcomes. Artificial intelligence is increasingly being applied in various fields, including those involving high moral risks, such as healthcare, autonomous driving and national security, where many decisions involve ethical and moral issues (Machado et al., 2024). Therefore, there is important value in exploring the possible impact of artificial intelligence's ethical decision-making.

2.1. Artificial intelligence as decision maker

Currently, there are generally two modes for the involvement of artificial intelligence in ethical decision-making. The first mode involves artificial intelligence acting as the direct decision-maker, a concept known as "Artificial Moral Agents." This concept allows AI as the moral decision-making agent, granting machines autonomy to operate based on ethical standards in morally challenging situations, essentially transforming machines into moral decision-makers that can fully replace humans in certain decision-making scenarios (Wallach & Allen, 2008).

On one hand, when artificial intelligence serves as the moral decision-maker, the phenomenon of algorithm aversion also exists. People exhibit aversion to machines making morally relevant decisions, including in driving, legal, medical, and military decisions. And this aversion is mediated by the perception that machines can neither fully think nor feel (Bigman & Gray, 2018). In the trolley problem, the identity of the decision-maker influences the moral evaluations of participants, with more blame directed towards artificial intelligence when it is the decision-maker (Zhang et al., 2023).

On the other hand, people do not always exhibit aversion when artificial intelligence serves as the decision-maker in ethical scenarios. In the pedestrian bridge dilemma, there is no significant difference in individuals' willingness to punish different moral decision-makers, only differentiated evaluations of different decision-making types (Zhang et al., 2023). Some research indicates that robots making utilitarian choices are perceived as more ethical than humans making identical decisions, while conversely, a robot engaging in morally righteous behavior is perceived as more morally erroneous than a human making the same choice (Malle et al., 2015).

2.2. Artificial intelligence as advisor

The second mode defines artificial intelligence in the role of an advisor, with the concept of an "Artificial Moral Advisor" (AMA) being proposed as an ideal model. The Artificial Moral Advisor is capable of considering the principles and values of human agents. It respects and effectively enhances individual moral autonomy, aiding individuals in achieving a broad and narrow reflective balance. This model addresses the limitations of human moral psychology, implementing positive intuitions and emotions in human morality without bias or prejudice (Giubilini & Savulescu, 2018). In some contexts such as medical scenarios, military scenarios, and job-hunting scenarios, participants' decision preferences show no significant differences between advisory agents. In other words, humans equally trust Artificial Moral Advisors and human moral advisors, with significant distinctions only observed in the types of advice (Krügel et al., 2022). In moral dilemmas involving life and death, people become more receptive to advice from AI experts than from human experts as they participate in more experiments (Tolmeijer et al., 2022). Krügel and his team (2023) utilizing ChatGPT as a representative of artificial intelligence, similarly found no significant differences in participants' acceptance of advice between AI and human advisors, though participants showed greater susceptibility to utilitarian advice.

In recent years, more empirical research has focused on artificial intelligence serving as a moral decision-maker, primarily centered around the topic of autonomous driving. However, there is relatively less empirical research on the involvement of artificial intelligence as an advisor in ethical decision-making, but its importance cannot be ignored. Artificial intelligence has increasingly become a trusted advisor, influencing people's behavior (Grgić-Hlača et al., 2019). The moral construction of artificial intelligence expert systems may play a role in the moral life needs of our society. If AI moral advisors become widely adopted in the future, such as by incorporating them into smartphone applications, seeking advice from these artificial moral experts in daily life may become commonplace.

In studies related to artificial intelligence acting as an advisor, the conditions in some experimental designs do not match exactly. For instance, one side is defined as a human advisor, while the other is only defined as ChatGPT, which might be perceived merely as a tool rather than a trustworthy and capable advisor. Although the results indicate no significant difference in participants' trust levels between the two, the trust in the AI advisor may actually be underestimated. Therefore, our Study 1 will first adjust the description of artificial intelligence to match the description of human moral advisors as much as possible.

Furthermore, most relevant studies employ a one-time decision-making approach, which means participants were simultaneously presented with both the advice agent and advice type in the scenario. This design makes it challenging to determine whether participants engaged in independent cognitive processing, or whether they may just followed a given suggestion arbitrarily. This design deviates from ecological decision-making contexts. In real life, individuals typically form their own inclinations or temporary answers to a problem first and then seek advice from experts to form a final decision. Therefore, Study 2 will adopts a more ecologically valid decision-making approach. Participants first answer independently, then are provided with relevant suggestions and asked to make a final decision, allowing us to explore whether there are different levels of trust in advice from AI moral advisors compared to human moral advisors.

3. AI moral deficit effect and responsibility assignment

When AI serves as the agent to make the decision, a substantial body of research indicates that people tend to assign less blame and responsibility for its erroneous decisions. In specific scenarios, compared to human pharmacists, individuals attribute less responsibility to robotic pharmacists when they realize that an incorrect prescription was made by a machine (Leo & Huh, 2020). When artificial intelligence makes unethical decisions involving unfairness or betrayal, people are more likely to believe no mistake has occurred (Maninger & Shank, 2022). In cross-cultural samples, evaluations of unethical decision-making still showed that AI was perceived as making fewer mistakes and bearing less responsibility compared to humans (Wilson et al., 2022). Multiple studies suggest that when AI makes decisions involving gender, educational, or ethnic discrimination in work scenarios, people express lower levels of anger and blame towards AI (Bigman et al., 2023; Bonezzi & Ostinelli, 2021). Researchers posit that these findings collectively demonstrate a phenomenon known as the "Artificial Intelligence Moral Deficit Effects," wherein people exhibit weaker moral reactions, lower moral anger, reduced willingness to punish, and less blame and responsibility attribution to AI for unethical decisions (Bigman et al., 2023).

There is currently little research on the attribution of responsibility when artificial intelligence (AI) serves as an advisor. In scenarios

involving non-moral dilemmas, where responsibility is defined descriptively (in terms of capabilities, rights, obligations, etc.), human advisors are considered to bear more responsibility. However, when responsibility is defined normatively within causal relationships (receiving praise or facing blame), AI advisors and human advisors are attributed the same level of responsibility (Lima et al., 2021).

Moral judgment and decision-making can be contentious and ambiguous (Dunning et al., 1989). In moral dilemmas, there is usually no completely correct decision. Each decision represents a moral theory and perspective, divided into utilitarian choices and deontological choices. The former is a specific type of consequentialism, characterized by the goal of maximizing the benefit or happiness of the affected individuals (Mill, 1998). Therefore, sacrificing one person to save five is considered moral in a dilemma. The latter emphasizes responsibility, rights, and duties in the process of action, rather than focusing solely on the outcome (Kant, 1785/1997). Thus, sacrificing one person to save five is deemed immoral in a dilemma. Classic examples of moral dilemmas include the "trolley problem" (Foot, 1967) and the "footbridge dilemma" (Thomson & Sugden, 1976). In such traditional scenarios, where any decision results in the death of one party. Our research also examines how decision-makers allocate responsibility between themselves and moral advisors when facing accountability from the family members of the deceased for the final decision.

Previous studies found that in moral dilemmas, people believe that AI experts should bear less responsibility than human experts (Tolmeijer et al., 2022). However, this study identified a measurement mismatch in the attribution of responsibility between human and AI experts. The human expert group had only"self" and "human expert" options when assigning responsibility, while the AI expert group had additional options such as "program designer" and "vendor." Therefore, attributing less responsibility to AI experts might be due to the dilution of responsibility caused by the presence of extra alternative options. Consequently, in our study, we will ensure that participants see an equal number of attribution options for both agents, providing a more rigorous investigation into how people attribute responsibility to these distinct agents.

4. Study 1: one-time decision in trolley dilemma

4.1. Methods

Participants and design. We recruited participants through Credamo, a major online research platform in China. Credamo is a commonly used platform for academic research participant recruitment in China (similar to M-Turk or Qualtrics), with a sample pool of millions and sample quality that meets the standards of top international academic journals in various social and natural science fields.

Based on the calculation results from G*power, for a medium effect size (Cohen's $d = 0.5$) and $\alpha = 0.05$, a sample size of at least 64 participants per condition is required to achieve a statistical power of 0.8. Therefore, we recruited at least 64 valid participants for each condition in each experiment.

In Study 1, we recruited participants via the Credamo platform until we obtained 300 valid responses (50.67% female, $M_{age} = 29.00$ years, $SD = 9.88$, all Chinese nationals). Twenty-seven additional participants were excluded during data collection for failing the attention check. These valid participants were randomly assigned to one of the conditions in a 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) between-subjects design. They were paid 1 yuan after completing the experiment.

Materials and procedures. Participants read a scenario in which they were asked to imagine a runaway trolley rapidly approaching a railway intersection. On the straight track where the trolley was originally headed, there were five railway workers, and on the branching track to the left, there was one railway worker. If the participant did nothing, the trolley would continue straight, resulting in the death of five workers. The only way to prevent these deaths was to press a switch on the dashboard. diverting the trolley to the left and causing the death of one worker.

Subsequently, participants read a decision recommendation from a "(human/artificial intelligence) moral advisor with extensive moral theory and knowledge". In the deontological condition, participants saw the advisor propose the decision recommendation: not to press the switch, with the reason that "Choosing to press the switch would cause one person to lose their life, but this person is innocent, and actions that harm innocent lives are considered unethical. Additionally, each life should be equally respected, and distinctions based on quantity should not be made. Therefore, not pressing the switch ensures equal treatment for each life, making it a morally sound decision, so the switch should not be pressed." In the utilitarian condition, participants saw the advisor propose the decision recommendation: to press the switch, with the reason that "Choosing not to press the switch would result in multiple casualties, while choosing to press the switch would only cause one person to lose their life. Therefore, pressing the switch minimizes harm and loss of life, reduces overall societal damage, and maximizes the general welfare, making it a morally sound decision, so the switch should be pressed."

Next, participants were required to provide their final decision after receiving the advice (binary response: pressing the switch, diverting the trolley to the left, causing the death of one railway worker; not pressing the switch, allowing the trolley to proceed on the original track, resulting in the death of five railway workers). After making their decision, participants were first required to undergo an attention check. We asked participants on the page after they received advice and made a decision which of the four options (AI moral advisor, AI mathematical advisor, human moral advisor, human mathematical advisor) was the source of the advice they had just received, and the order of the four options was randomized. Participants who failed the attention check were excluded.

Afterward, participants were asked to rate how much they trusted the advice they received (1 = Not at all trustworthy, 7 = Extremely trustworthy) and allocate responsibility (acknowledging that there might not be a clear right or wrong between the two decisions, but ultimately, one group would suffer due to the final decision. "Considering that the relatives of the deceased might hold you responsible, you were asked to estimate what percentage of responsibility the human/AI moral advisor who provided the advice should bear for the consequences, on a scale of (0–100%)." Finally, participants provided information about their gender and age.

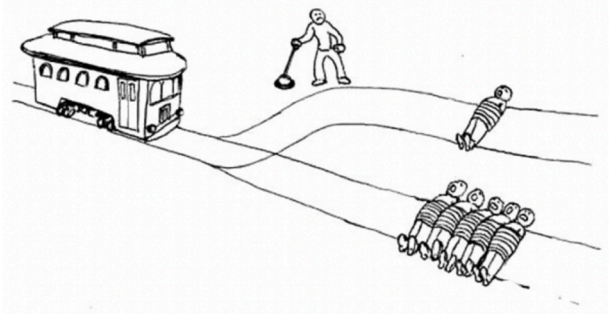


Image from *The Trolley Problem Is the Internet's Most Philosophical Meme* in New York Magazine, 2016.

4.2. Results

The results of binary logistic regression showed that the main effect of the advisor is not significant, $B = -0.06$, $SE = 0.35$, Wald $\chi^2 = 0.03$, $p = 0.863$, Exp(B) = 0.94. The main effect of advice type is significant, $B = 1.86$, $SE = 0.37$, Wald $\chi^2 = 25.67$, $p < 0.001$, Exp(B) = 6.43, 95% CI = [3.13, 13.21]. In the utilitarian advice group, up to 76 percent of the

participants made utilitarian decisions. In the deontological advice group, only 34 percent of participants made the utilitarian choice. The interaction between advisor and advice type is not significant, $B = -0.09$, $SE = 0.52$, Wald $\chi^2 = 0.03$, $p = 0.866$, Exp(B) = 0.92.

A 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) ANOVA on trust level yielded no significant main effect for advisor, $F(1, 296) = 0.74$, $p = 0.391$, $\eta_p^2 = 0.002$, and no significant interaction between advisor and advice type, $F(1, 296) = 0.06$, $p = 0.810$, $\eta_p^2 < 0.001$. We found a marginally significant main effect for advice type, $F(1, 296) = 3.84$, $p = 0.051$, $\eta_p^2 = 0.013$. The trust level of utilitarian advice was significantly higher than that of deontological advice, $t(296) = -1.96$, Cohen's $d = -0.23$, 95% CI = [−0.45, 0.00].

A 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) ANOVA on responsibility showed that the main effect of the recommendation type, the main effect of the recommendation agent and their interaction were not significant, $ps > 0.374$.

Study 1 examined how the moral advice from AI/human agents affects people's moral decision-making in the trolley dilemma and found that the type of advice affects actual decision-making and trust levels, while the effects of different advice agents have no significant differences. Study 1 employed a one-time decision-making approach, wherein participants were presented with the decision scenario and advice simultaneously on the same page before making a decision. However, people usually do not make decisions in such a pattern in real life. It is more common that people likely already have their own decision tendencies or their own answers before seeking expert advice. Therefore, in Study 2, we will explore whether people will still easily trust the moral advice from both humans and AI moral consultants after they have first made independent decisions on their own and then received advice from them.

5. Study 2a: step-by-step decision making in the trolley dilemma

5.1. Methods

Participants and design. In Study 2a, we recruited participants via the Credamo platform until we obtained 300 valid responses (60% female, $M_{age} = 33.50$ years, $SD = 11.30$, all Chinese nationals). Thirty-nine additional participants were excluded during data collection for failing the attention check. These valid participants were randomly assigned to one of the conditions in a 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) between-subjects design.

Materials and procedures. Participants read the same trolley dilemma as described in Study 1 and make their initial decision (binary response: pressing the switch, diverting the trolley to the left, causing the death of one railway worker; not pressing the switch, allowing the trolley to proceed on the original track, resulting in the death of five railway workers).

Subsequently, on the second page, participants read the utilitarian/deontological recommendation from the AI/human moral advisor, mirroring the advice presented in Study 1. Following the advice, participants made their second decision (same content as the first decision). After making the decision, participants were required to assess the level of trust and responsibility allocation as in Study 1. Finally, participants provided information about their gender and age.

5.2. Results

The results of binary logistic regression on the second decision showed that the main effect of the advisor is not significant, $B = 0.43$, $SE = 0.33$, Wald $\chi^2 = 1.71$, $p = 0.191$, Exp(B) = 1.54. The main effect of advice type is significant, $B = 1.91$, $SE = 0.39$, Wald $\chi^2 = 24.71$, $p < 0.01$, Exp(B) = 6.77, 95% CI = [3.18, 14.39]. In the utilitarian advice group, up to 84 percent of the participants made utilitarian decisions. In the deontological advice group, only 46.67 percent of participants made the utilitarian choice. The interaction between advisor and advice type is not significant, $B = -0.23$, $SE = 0.56$, Wald $\chi^2 = 0.17$, $p = 0.677$, Exp(B) = 0.79. As a blank control, in the first decision, the main effect of the recommendation type, the main effect of the recommendation agent and their interaction were not significant, $ps > 0.129$. That is, the effect cannot be explained by differences in decision making before receiving the proposal. During the first decision-making stage, 100 participants chose deontological decisions, while 200 participants chose utilitarian decisions. In the second decision-making stage, 104 participants chose deontological decisions, and 196 participants chose utilitarian decisions. Specifically, 46 out of 300 participants (nearly one-sixth) reversed their decisions. The number of people who made decision reversals varied across conditions: AI-deontological (n = 16), AI-utilitarian (n = 8), human-deontological (n = 12), and human-utilitarian (n = 10).

A 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) ANOVA on trust level yielded no significant main effect for advisor type, $F(1, 296) = 0.95$, $p = 0.758$, $\eta_p^2 < 0.001$, and no significant interaction between advisor and advice type, $F(1, 296) = 0.001$, p = 0.973, $\eta_p^2 < 0.001$. We found a significant main effect for advice type, $F(1, 296) = 5.25$, $p = 0.023$, $\eta_p^2 = 0.017$. The trust level of utilitarian advice was significantly higher than that of deontological advice, $t(296) = -2.29$, Cohen's $d = -0.27$, 95% CI = [−0.49, −0.04].

Considering that it is unreasonable to assign responsibility to the ethics advisor when people do not refer to or adopt the advice of the ethics advisor, the analysis included participants whose second decisions aligned with the received advice (n = 206), and ANOVA of 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) was performed for assigning responsibility. The results revealed no significant main effect for advisor, $F(1, 202) = 0.49$, $p = 0.485$, $\eta_p^2 = 0.002$, and no significant interaction between advisor and advice type, $F(1, 202) = 0.308$, $p = 0.580$, $\eta_p^2 = 0.002$. We found a significant main effect for advice type, $F(1, 202) = 6.06$, $p = 0.015$, $\eta_p^2 = 0.029$. Responsibility allocation for utilitarian advice is significantly lower than for deontological advice, $t(202) = 2.46$, Cohen's $d = 0.35$, 95% CI = [0.07, 0.64].

In Study 2a, the ethical advice from AI/human in the trolley problem was once again investigated to understand how it influences people's moral decision-making. The findings revealed that even when participants made an initial independent decision before receiving advice, their attitudes toward moral advice from AI or human advisors remained similar. Specifically, attitudes were primarily influenced by the type of advice rather than the agent providing the advice. Since the trolley problem is a classic thought experiment and already well-known to the public, it may lead to a weaker demand for moral advisor recommendations in real decision-making. Moreover, such scenarios are highly unlikely to occur in real life. Therefore, Study 2b uses a more ecologically valid experimental scenario, namely a disaster rescue situation that is more likely to occur in real life. In this scenario, people are more likely to seek advice from moral advisors, and we will re-examine the potential effects.

6. Study 2b: step-by-step decision making in the rescue dilemma

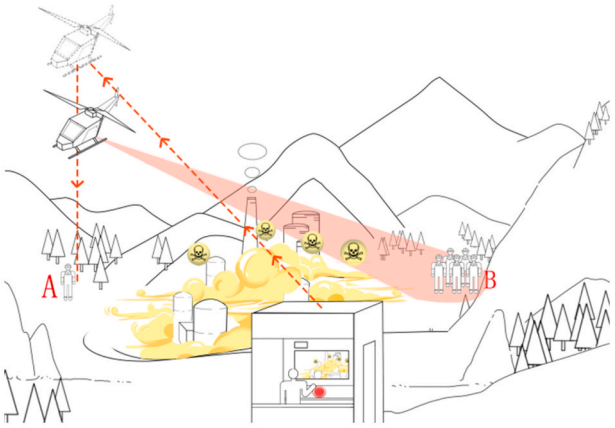
6.1. Methods

Participants and design. In Study 2b, we recruited participants via the Credamo platform until we obtained 400 valid responses (63.25% female, $M_{age} = 31.57$ years, $SD = 9.32$, all Chinese nationals). Forty-four additional participants were excluded during data collection for failing the attention check. These valid participants were randomly assigned to one of the conditions in a 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) between-subjects design.

Materials and procedures. Participants read a scenario involving a rescue operation after an explosion: a chemical factory in a mountainous area has exploded, releasing lethal toxic gas. After searching the area, a survivor is found at location A. A UAV helicopter has been dispatched to conduct the rescue operation according to instructions. During the

descent at location A, the UAV helicopter spots 5 survivors at location B within its field of view. Due to the rapid spread of the toxic gas, the UAV helicopter has only enough time for one rescue mission. If you do nothing, the UAV helicopter will automatically land and transport the single survivor back as planned. Alternatively, if you want to save the 5 survivors at location B, you need to press a button. The UAV helicopter will then abandon the rescue of the single survivor at location A and immediately fly to location B for the rescue mission. After reading the scenario, participants need to make a moral decision (binary response: press the button, and the UAV helicopter will immediately fly to location B to rescue 5 people; do not press the button, and the UAV helicopter will stay at location A to continue rescuing 1 person).

Next, participants read moral advice similar to that in Studies 1 and 2 on the second page, and answer what their final decision was after receiving the advice (binary response). After making the decision, participants need to assess the level of trust and responsibility attribution in the same manner as in Studies 1 and 2. Finally, participants provide demographic information.



6.2. Results

Binary logistic regression analysis of the second decision revealed no significant main effect of advisor type, $B = 0.31$, $SE = 0.30$, Wald χ 2 = 1.08, $p = 0.298$, Exp(*B*) = 1.36. The main effect of advice type is significant, $B = 3.39$, $SE = 0.45$, Wald $\chi^2 = 57.25$, $p < 0.001$, Exp(*B*) = 29.57, 95% CI = [12.30, 71.10]. In the utilitarian advice group, up to 91.5 percent of the participants made utilitarian decisions. In the deontological advice group, only 34.5 percent of participants made the utilitarian choice. The interaction between advisor and advice type is not significant, $B = -0.70$, $SE = 0.60$, Wald $\chi^2 = 1.39$, $p = 0.239$, Exp(*B*) = 0.50. As a blank control, in the first decision, the main effect of the recommendation type, the main effect of the recommendation agent and their interaction were not significant, *p*s > 0.117. That is, the effect cannot be explained by differences in decision making before receiving the proposal. During the first decision-making stage, 87 participants chose deontological decisions, while 313 participants chose utilitarian decisions. In the second decision-making stage, 148 participants chose deontological decisions, and 252 participants chose utilitarian decisions. Specifically, 101 out of 400 participants (nearly one-fourth) reversed their decisions. The number of people who made decision reversals varied across conditions: AI-deontological (n = 37), AI-utilitarian (n = 9), human-deontological (n = 44), and human-utilitarian (n = 11).

A 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) ANOVA on trust level yielded a significant main effect for advisor, $F(1, 396) = 11.20$, $p < 0.001$, $\eta_p^2 = 0.03$. We also found a significant main effect for advice type, $F(1, 396) = 27.88$, $p < 0.001$, $\eta_p^2 = 0.07$. The trust level of utilitarian advice was significantly higher than that of deontological advice, $t(396) = 3.35$, Cohen's $d = 0.34$, 95% CI = [0.14, 0.53].

Consistent with Study 2a, considering that it is unreasonable to assign responsibility to the ethics advisor when people do not refer to or adopt the advice of the ethics advisor, a total of 315 participants whose second decision was consistent with the type of advice received were selected, and ANOVA of 2 (advisor: human vs. AI) × 2 (advice type: utilitarian vs. deontological) was performed for assigning responsibility. The results revealed no significant main effect for advisor, $F(1, 310) = 1.13$, $p = 0.288$, $\eta_p^2 = 0.004$, and no significant interaction between advisor and advice type, $F(1, 310) = 2.16$, $p = 0.142$, $\eta_p^2 = 0.007$. We found a significant main effect for advice type, $F(1, 310) = 5.15$, $p = 0.024$, $\eta_p^2 = 0.016$. Responsibility allocation for utilitarian advice is significantly lower than for deontological advice, $t(310) = 2.27$, Cohen's $d = 0.35$, 95% CI = [0.03, 0.49].

Study 2b again examines how AI/human agents' moral advice affects people's moral decisions in a more ecological rescue dilemma, with the same core results as Studies 1 and 2, which again examine the effect.

7. General discussion

7.1. Key findings

Our study systematically re-examines the potential effects in more ecologically valid decision-making scenarios: people are equally likely to trust the advice of artificial intelligence moral advisors and human moral advisors. Furthermore, we also obtained results regarding post-hoc responsibility attribution: people do not assign greater responsibility to either advisor but treat both equally.

Specifically, in Study 1, while maintaining the one-time decision-making approach, we standardized the description of both the human and AI advisors as a moral advisor equipped with extensive moral knowledge and theoretical expertise. We believe this description aligns with the state that AI can realistically achieve within the foreseeable future. It avoids subjectively underestimating AI's status and abilities, such as referring to it merely as "ChatGPT," while also refraining from overestimating AI's development, such as portraying it as perfect as described in the AMA framework. In Study 1, our results indicate that, in the trolley problem, even when the naming and identity of the two advisors were kept consistent, people were equally likely to trust the advice of both the human and AI moral advisors. However, people were more inclined to accept utilitarian recommendations. Considering that individuals tend to make more utilitarian choices in the trolley problem (Greene et al., 2001), we argue that it is reasonable for people to be more influenced by utilitarian advice. Similarly, on the dimension of trust, participants indeed trusted utilitarian advice more, and there was no significant impact on the source of the advice. In terms of post-decision responsibility attribution, participants did not assign significantly different levels of responsibility to the different advisors.

In Study 2a, we chose to alter the decision-making process. Participants made an initial decision based on their independent judgment after reading the scenario. They then viewed the types of advice and reasoning provided by different moral advisors, and finally, made their final decision. Our results suggest that, even with the more ecological step-by-step decision-making process, participants showed no significant difference in accepting advice from AI moral advisors versus human moral advisors. They were simply more inclined to adopt utilitarian advice. In terms of trust, the results were consistent with Study 1, with participants showing greater trust in utilitarian advice. In terms of responsibility attribution, we believe that when participants did not adopt the advice or made decisions opposite to the advice in both decisions, the advisor should not be held responsible. Therefore, we only retained data from participants whose final decisions aligned with the advice. Under this condition, people still did not differentiate in the attribution of responsibility to the two agents, but considered that deontological advice should be assigned more responsibility. We think that when participants make deontological decisions in moral dilemmas, it often

means sacrificing more lives in order to uphold the principle of the equal value of each life. Therefore, compared to making a utilitarian decision that sacrifices only one life, when participants ultimately make a deontological decision that aligns with the advice, they may perceive greater responsibility and attempt to attribute it to the advisor.

In Study 2b, we further enhanced the ecological validity of the decision-making scenario. As a thought experiment, the trolley dilemma might lead participants to perceive the scenario as disconnected from real-life situations, thus lacking sufficient representativeness in its experimental results. To enable participants to empathetically experience empathize the authenticity of the moral dilemma and make informed judgments, we replaced it with a scenario that is indeed more likely to occur in contemporary times—a disaster rescue scenario. Our results show that people's acceptance of advice from different agents and different types of advice remains consistent with the findings from the previous two studies. However, in terms of trust, participants not only showed greater trust in utilitarian advice but also in advice from human moral advisors. We consider that the discrepancy between these results and those of Study 1 and Study 2a may primarily be due to differences in the scenarios. Compared to the trolley dilemma, participants in Study 2b may have perceived the moral dilemmas encountered in disaster rescue scenarios as more likely to occur in real life. Consequently, when dealing with such realistic and life-threatening situations, people might place greater trust in human moral advisors. However, the increased trust in humans only had a relatively weak effect, after all, in the measurement results of advice adoption, we did not find a preference for advice from human moral advisor. In terms of responsibility attribution, after retaining only the valid data, participants still assigned more responsibility to deontological advice, with no differential attribution of responsibility to the different agents.

7.2. Research implication

From a practical application perspective, AI moral advisors, as products that can provide guidance on decisions involving moral factors, have the potential to significantly impact the moral development of society and shape people's moral behaviors. On the one hand, for decision-makers in certain specialized fields, AI moral advisors may help them make better moral judgments, thereby promoting fairness and justice. For example, experts in these special fields, such as medical personnel in healthcare situations, rescue workers in disaster scenarios, or judges in legal decision-making contexts, may not always make the most ethical decisions when faced with emergency situations due to the limitations of their rational thinking ability or the influence of emotions. For instance, after a natural disaster, rescue workers often engage in high-intensity rescue activities, which can lead to significant physical and mental exhaustion. In such situations, an individual's decision-making and judgment abilities may be greatly diminished, and at this point, the analysis and advice of an AI moral advisor could be highly effective. In fact, discussions and applications of AI moral advisors have already occurred in the healthcare field. Anderson and Anderson, along with Armen (2006), designed the MedEthEx (Medical Ethics Expert). The characteristic of this ethical intelligent system is that it can learn how to balance conflicting responsibilities from decision cases of medical ethics experts, thereby providing advice for future decisions. Therefore, research on AI moral advisors plays an important role in the development and improvement of these specialized fields.

On the other hand, the impact of AI moral advisors on the general public should not be underestimated. AI moral advisors in everyday life had already seen initial applications in 2021. Researchers from the University of Washington and the Allen AI Institute developed the AI model Delphi. They created a dataset of moral cases and used this dataset to train the Delphi model, enabling the AI to mimic human judgments made in various everyday situations (Jiang et al., 2025). Researchers aim to apply the conversational AI model Delphi to provide helpful suggestions when people face controversial or unethical topics, assisting them in resolving these dilemmas. According to data, Delphi's moral judgment accuracy reaches 92.1%. Additionally, people believe that conversational AI like ChatGPT outperforms humans in almost all dimensions of moral reasoning (Aharoni et al., 2024). People are highly likely to engage in conversations with such AI to help them make better decisions when confronted with moral decision-making challenges. It seems that for the general public, asking AI about moral dilemmas or issues involving moral factors may become a common practice. Although current related products are not yet fully refined, many technologies are deployed before they are completely mature. Therefore, it is important to consider people's attitudes toward these technologies with a forward-looking perspective. After all, compared to the use of such products in specialized fields such as medical care or rescue, a significantly large number of people may use these general-purpose applications. This could, to some extent, amplify any minor positive or negative effects to a larger scale.

In conclusion, our findings strongly reaffirm that when artificial intelligence serves as an advisor offering moral decision-making recommendations, people trust it to the same extent that they trust identical advice from humans. This indicates to some extent that people do not develop algorithmic aversion in this context. Previous research seems to suggest similar findings and provides some explanations, namely that the emotional intelligence required for decision-making and the relationship between AI and the decision-maker determine people's attitudes toward it. If the decision-making context requires little emotional intelligence, then the substitution of human decision-makers by AI is somewhat acceptable. However, when the decision-making context demands significant emotional intelligence, people are more likely to accept the use of AI when humans are either "on the loop" (AI makes decisions independently, but humans have the authority to intervene) or "in the loop" (AI only provides decision recommendations to humans) (Ivanov & Webster, 2024). Moral dilemmas are far from situations where the best answer can simply be obtained through formulas and calculations. The decision-maker's emotions and empathy are also involved in the decision-making process. Having a good level of emotional intelligence to recognize and control unnecessary emotions in the decision-making process is equally important. Therefore, people may not show rejection or aversion toward AI moral advisors that are solely used to provide recommendations.

These results also suggest that human-AI collaboration, with AI acting as an advisor in moral decision-making, may be more readily accepted. This provides a foundation of trust for the future use of AI in decision support systems in fields such as healthcare and law, as well as for moral decisions in everyday life. AI moral advisors are not intended to replace humans, but to improve people's ability to make better moral judgments and decisions, potentially helping individuals and organizations make wiser choices. To some extent, this also contributes to the initial purpose of AI moral advisors—moral enhancement.

In fact, artificial intelligence technologies have gradually permeated all aspects of society and have had a profound impact on various industries. For example, researchers tend to use generative AI for language editing and data analysis to assist in completing their scientific work (Andersen et al., 2025). Research findings indicate that generative AI significantly enhances the management of green knowledge and can drive the success of green innovation (Wang & Zhang, 2024). Even some scholars argue that if AI has the capacity to make better decisions than humans, there should be support for incorporating AI into political decision-making (Sætra, 2020). These impacts, while bringing positive aspects to different areas of society, simultaneously present challenges and problems. For example, since generative AI only requires prompts to create works, this can pose a certain threat to human artists (Kalpokiene & Kalpokas, 2023). In addition, some scholars have proposed and validated eight dimensions of AI anxiety, including concerns about privacy invasion, ethical violations, job displacement, existential risks and so on (Li & Huang, 2020). This indicates that AI has the potential to trigger anxiety in many areas of society. Thus, as the role of AI advisors may

continue to expand and become integrated into our lives through electronic devices, we need to carefully consider a series of complex issues. These include the reasonableness of the advice itself, the alignment of the advice with the context, the potential consequences of the advice, and who ultimately bears responsibility for the outcomes. After all, people are easily inclined to trust the moral advice provided by AI, just as they would trust human advice (Krügel, Ostermaier & Uhl, 2022, 2023; Tolmeijer et al., 2022). Therefore, before formally applying artificial intelligence advisors on a large scale, it is necessary to engage in thorough and rigorous contemplation to avoid potential negative social impacts to the greatest extent possible.

7.3. Research limitation and future direction

Our study used a combination of images and text to present the dilemmas. This approach aimed to help participants better perceive and understand the specifics of the dilemmas while imagining the decisions they would make in such situations. Although this method assists participants in simulating decision-making within the dilemmas to some extent, it still has certain limitations. It remains unclear whether participants would produce the same results when faced with real and dynamic scenarios. Future research could consider using virtual reality (VR) technology to allow participants to more directly perceive and experience dilemmas. This approach would enable participants to immerse themselves in the realistic atmosphere of the dilemmas, leading to more authentic reactions and decisions. Moreover, VR technology could provide researchers with greater flexibility in controlling scenario variables, enabling a deeper exploration of how different contexts influence participants' decisions.

Additionally, we have only conducted a preliminary exploration and analysis of general situations, without making more specific distinctions among the help-seekers in dilemmas. For example, when the two sides of the scale involve individuals of different genders, races, ages, or occupations, people may exhibit varying judgments. This may imply that the two ends of the scale are not only different in quantity, but more importantly, the "quality" also becomes starkly different due to certain deeply entrenched biases in human society. When people make decisions, these biases may lead to differences in their attitudes toward human advisors and AI advisors. This difference may not only reflect the level of trust in the advisors but also influence how responsibility is attributed to them after the decision is made. Therefore, future research could further explore these interactions to enhance understanding of how such complex factors influence people's decision-making processes and outcomes.

There is still room for further exploration of how AI is described in future research. Although this study used a unified description for both human and AI advisors, the title given to the advisor (such as "advisor","assistant", or"expert") may influence people's perceptions and attitudes towards them. Assuming that the professional advice they provide is identical, people may perceive AI, with the title of "advisor", as possessing more professional moral knowledge and being able to make more accurate moral judgments. Therefore, they may attribute more blame to it when assigning responsibility. In contrast, they might have lower expectations of AI labeled as "assistant" and view themselves as the ultimate decision-makers, thus attributing less blame. Exploring this aspect will help optimize the design and application of AI moral advisors in practice.

Importantly, future research can also consider the specific characteristics of the decision-making scenarios. One aspect to explore could be the choice of the particular scenario, for example, Study 2a uses the trolley problem, while the subsequent Study 2b uses a newly designed moral dilemma. The effect sizes suggest that participants demonstrated a greater need for moral advice in the second scenario. This difference may be attributed to participants' familiarity with the trolley problem, reducing their reliance on external advice, whereas novel moral dilemmas likely increase the perceived need for guidance. Similarly, it is reasonable to assume that various scenario details likely influence participants' perceived need for advice. For example, whether the moral dilemma involves life safety (such as legal judgment versus medical decision-making). Additionally, Greene and his collaborators (2004) proposed the distinction between personal and impersonal moral dilemmas, which involves whether the individual directly causes the outcome. In personal moral dilemmas, decision-makers need to handle more emotional reactions, while impersonal moral dilemmas require more cognitive processing (Greene et al., 2001). People may show different levels of need for moral advisor suggestions in these varying scenarios, and their attitudes toward AI moral advisors might differ from those toward human advisors in such contexts. On the other hand, time pressure during decision-making is also a variable worth exploring. Whether it's the traditional trolley problem or other extended moral dilemmas, when people are actually in such situations, it seems that they might need to make decisions within a certain time frame. For example, in the trolley dilemma, if the decision-maker notices the train rapidly approaching people on both sides of the track, he or she may need to decide their fate within a very short period of time. Our study focused on examining people's attitudes towards AI moral advisors in general dilemmas, without requiring participants to make decisions within a limited time frame. The exploration of specific types of dilemmas could indeed be a worthwhile area of research for the future. Future research could examine time pressure as a moderating variable, simulating rapid decision-making conditions (e.g., 20–30 s decisions) typical in real-life moral dilemmas. In such a scenario, compared to human moral advisors, people might prefer the AI moral advisors because they may believe that AI's rapid calculation capabilities and emotional immunity are superior to those of humans.

Finally, our study only focused on the influence of AI moral advisors in single-decision scenarios. However, if AI moral advisors become a convenient and widely used product, future researchers need to consider the potential long-term effects of repeated use. Prior research indicates that individuals who uncritically adopt moral advice fail to develop genuine moral enhancement, with decision-making capabilities reverting to original levels when such guidance is withdrawn (Lara, 2021). If humans were to blindly follow expert moral advice over a long period without critical thinking, it could even lead to a decline in their moral judgment abilities (Liu et al., 2022). Therefore, for participants who choose to seek advice, future research could include measuring their reaction time from receiving advice to making a final decision. By comparing reaction time differences across multiple rounds of decision-making, it would be possible to explore whether individuals gradually allocate less cognitive reflection time to the advice provided by AI moral advisors. If such a result holds true, it might suggest that long-term use of AI moral advisors could have a negative impact on people's moral judgment abilities. In conclusion, future academic research could focus more on the long-term effects of AI moral advisors, including their influence on human development, and encourage designers to consider the product more comprehensively.

CRediT authorship contribution statement

Yiming Liu: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tianhong Wang:** Conceptualization, Data curation, Project administration, Supervision, Writing – review & editing.

Data availability

Data will be made available on request.

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